Question: What is the function of a **virtual machine manager**?

1) Sends resource requests from multiple virtual machines to the hypervisor kernel
2) Abstracts physical hardware and presents It to a virtual machine
3) Maps the virtual machine hardware to the hypervisor kernel
4) Presents resource requests from a virtual machine directly to physical hardware

---

Question: What is a characteristic **of software-defined networking**?

1) Abstracts and separates the control plane from the data plane
2) Controller in network switches takes over data plane functions
3) Data plane sends instructions to the control plane to handle network traffic
4) External software combines both data plane and control plane functions

---

Question: In a VMware environment **the .vmdk** file represents a virtual machine's

1) virtual Disk
2) Virtual Memory
3) configuration file
4) none

Question: Which type of **hypervisor** is shown in the following figure?

Guest OS Guest OS
Hypervisor
System Hardware

1) Type 1
2) Type 2
3) Bare Metal
4) Both Type 1 and Bare Metal

Question: A customer in **an infrastructure as a service**

1) manages or controls the underlying cloud Infrastructure and has control over operating systems
2) does not manage or control the underlying cloud Infrastructure but has control over operating systems
3) manages and controls the underlying cloud Infrastructure but has no control over operating systems
4) does not manage or control the underlying cloud Infrastructure and has no control over operating systems

---

Question: Which of the following should be placed in **second lowermost layer** for the following figure

Guest OS Guest OS
Hypervisor
?
System Hardware

Answer: **Host Operating System**

Question: VM can be altered and if those changes prove to cause problems, the VM is reverted to the **original state** through the use of

1) Template
2) OVF Template
3) Clone
4) Snapshots

Question: VM **can not be powered on** and are used **to create new VMs** are known as?

1) Template 2) Clone 3) Snapshots

Question: All the below are true about **a cloned VM** except

1) Can be created from s running vm
2) Can be created manually
3) Requires customization to ensure uniqueness
4) Is affected by changes happening In the parent VM

Question: Which of the following is true about **DAS**

1) It is a simpler approach to storage that Is directly connected to the host computer
2) An external hard drive Is the most basic DAS system
3) Large DAS systems Include multiple SSDs/HODs
4) All mentioned

Question: A **Full Clone** requires ..... disk space that the original VM

1) Same amount of
2) Greater amount of A
3) less amount of

Question: What is a **hypervisor**?

1) Hardware that provides a virtualization layer on a computer system to create Virtual machines
2) Software that provides Remote access to existing virtual machines.
3) Software that manages Virtual Storage:
4) Software that provides 4 virtualization layer on a computer system to create VMs

---

Question: One of the advantages of **application virtualization** is

1) Complex OS image management
2) Remote access anywhere anytime
3) Resolve OS conflicts and compatibility Issues
4) Delegated application management

Question: What is an accurate statement about **a hybrid cloud**?

1) Allows an organization to outsource the Implementation of a private cloud to a cloud provider
2) Allows organizations with common concerns to share the cost of deploying the cloud
3) Enables the entire cloud Infrastructure to be controlled by the consumer's IT staff
4) Supports data and application portability for load balancing between clouds

---

Question: **Best-of-breed infrastructure** refers to

1) integrate the best-of-breed Infrastructure components (hardware and software) purchased from multiple different vendors
2) integrate the best-of-breed hardware components purchased from multiple different vendors
3) choose the best-of-breed Infrastructure components (hardware and software) purchased between multiple different vendors
4) integrate the best-of-breed infrastructure components (hardware and software) purchased from one vendor

---

Question: if the number of disks required for performance is 20.5 and capacity 10, how many disks should be bought

1) 20.5
2) 30.5
3) 21
4) 31
5) 10

Question: In **mirroring** the number of drives required is

1) Double the required amount of data
2) Same amount
3) An even number of drives
4) both (even and double required)

Question: In **RAID** 6 the write penalty is

1) 2
2) 3
3) 4
4) 5

Question: The **disk controller**, translates linear address of blocks into

1) A hard disk location defined hy Cylinder', head, sector
2) sequential partitions
3) Logical addressing on the disk
4) None of the above

Question: **SSDs** compared with **HDDs**

1) Have a response time faster than HDD
2) Have a response time slower than HDD
3) Best at long term activities
4) Cannot read randomly

Question: **Data Service time** includes

1) waiting time + rotational latency + data transfer time
2) seek time + rotational latency + data transfer time
3) waiting time + seek time + rotational latency + data transfer time
4) response time + seek time + rotational latency + data transfer time

---

Question: Compared with **RAID** 5

1) Lower cost
2) Same cost
3) Higher cost
4) None of the above

Question: The use of striped array With independent disks and dedicated **parity disk** is considered

1) RAID 1
2) RAID 2
3) RAID 3

4) RAID 4

Question: Using a dedicated **parity bit** in **RAID**.

1) Decreases disk speed
2) increases disk speed
3) increases disk efficiency
4) Decreases disk reliability

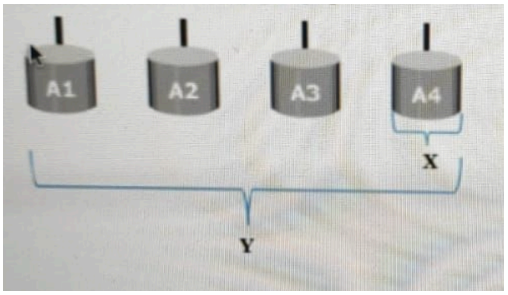
Question: If a system has a **seek time** of 1ms, **Rotational latency** of 5ms and **data transfer time of** 0.2ms, How many IOPS can the disk performance? System is response time sensitive

1) 120 10Ps
2) 112 10Ps
3) 100 10Ps
4) 91 10Ps

Question: the **hard disk head** read data by detecting the

1) the magnetic resonance
2) the magnetic polarization
3) the magnetic field strength
4) the magnetic field

Question: What's X and Y?



1) both are **strip**
2) both are **stripe**
3) X: strip Y: Stripe
4) X: stripe Y: Strip

Question: Given that the **average seek time** is equal to 12 ms. Calculate the **full stroke** time.

1) 6ms
2) 4 ms
3) 12ms
4) 36ms **12*3=36**

Question: Consider an application that generates 1220 IOPS at peak workload, with read/write ratio of 3:1. calculate **disk work** load at peak activity for **RAID** 3

1) 2100
2) 1300
3) 2500
4) 1700

Question: if the Application writes is 40% and 60% reads which raid level should be used for protection

1) **RAID** 0
2) RAID 1
3) RAID 5
4) RAID 6

Question: use of **RAID** stripping

1) increases efficiency
2) has no effect on efficiency
3) decreases efficiency
4) none

Question: **RAID** 6 can support

1) 1 disk failure
2) 2 disk failure
3) 3 disk failure
4) both 1&2

Question: Consider a system **utilization** 90% calculate the **maximum number of IOPS** a disk can perform with average **seek time** 3 ms, **rotational latency** 5 ms and **data transfer time** 2 ms.

1) 100 IOPs
2) 70 IOPs
3) 90 IOPS
4) 80 IOPs

Question: Calculate the **rotational latency** given that drive option 100 rps

1) 5 ms **0.5*1000/rps=5ms**
2) 0.5ms
3) 3 ms
4) 30 ms

Question: In cache Management Most Recently used an algorithm that

1) Is based on the assumption that recently accessed data may not be required for a while
2) is based On the assumption that least accessed data may not be required for a while
3) the pages that have been accessed least recently are freed up or marked for reuse
4) the pages that have been accessed most recently are freed up or marked for reuse

---

Question: What cache utilization level at which the storage system starts high-speed flushing of cache data

1) High water mark flushing
2) Medium water mark flushing
3) Low water mark flushing
4) Forced flushing

Question: **Vault drives** are

1) a set of physical storage drives to dump the contents of cache during power restore
2) a set of physical storage drives to dump the contents of cache during power failure
3) a set of physical storage drives to dump the contents of cache HWE
4) none of the above

Question: **LUN** created using virtual provisioning Is called
1) Virtual LUN
2) Thick LUN
3) LUN expansion
4) Thin LUN

Question: A disadvantage of scale-up **NAS** is
1) expensive
2) Requires multiple storage units
3) Complex
4) degrades performance

Question: Clients in the Scale out **NAS**
1) Connects to the NAS over a Switch
2) Access the entire system through one node
3) Has access to only one node
4) None of the above

Question: The internal Network in the Scale-out **NAS**
1) Connects external clients to the NAS.
2) Gives access to file sharing for users
3) Both 1& 2
4) connections for Intra-cluster communication

Question: **ANAS** system consists of
1) Controller
2) Storage
3) Both 1 & 2
4) None of the above

Question: The storage of the Gateway **NAS** Is:
1) Directly attached to the NAS Head
2) Managed by the NAS head
3) Accessed externally through a SAN Network
4) None of the above

Question: Data Is placed In the cache and immediately written to the storage, and an acknowledgment Is sent to the compute system Is called
1) send Through cache
2) Write back Cache

Question: Pick odd one out
a) Mirroring
b) Striping
c) Error-correction
d) Fault tolerance

Question: Which one of the following is not an advantage of **RAID**?

a) Data Security
b) Increased & integrated capacity
c) Improved performance
d) Effective data capacity increases

Question: Pick the false statement about **RAID** 0
a) Provides no redundancy
b) Provides data striping
c) improves performance
d) Provides fault tolerance

Question: Pick the wrong statement about **the hard disk**?

a) Hard disk has multiple platters and each platter has two read/write heads one on each side
b) It's a non-volatile & random-access storage device
c) Hard disk can only have IDE or USB interface
d) Data Transfer rate is over 80 MBPS

Question: Which of the following statements about various **hard disks** is wrong?

a) SATA Disks support faster transfer rates and have support for hot swapping
b) USB Hard disks store data on flash memory
c) ATA hard disks cannot be connected externally to computer
d) None of the mentioned

Question: Which of the following is false?

a) NVRAM has built in battery which keeps power applied to it even after the power is switched off
b) Flash memory can be electrically erased and reprogrammed
c) Flash memory is used in memory sticks
d) Flash disk can only have USB interface

Question: Which of the following is not true about **JBOD**?
a) JBOD can combine hard disks of different sizes into a single unit without loss of any capacity
b) If a drive in a JBOD set dies then it may be easier to recover the files on the other Drives
c) JBOD supports data redundancy
d) JBOD doesn't has any storage controller intelligence

Question: Which of the following is not a non volatile **storage device**?
a) Memory Stick
b) Hard Disk
c) Random Access Memory
d) NVRAM

Question: Which of the following is sequential access **storage device**?
a) Hard Disk
b) CD-ROM
c) Tape Cartridge
d) Main Memory

Question: Which of the Following is not an off-line **storage device**?

a) Tape Cartridge
b) Flash Memory
c) Tape Library
d) CD-ROM

Question: Which of the following **RAID** levels are nested?

a) Raid 5S
b) Raid 6
c) Raid 10
d) Raid 4

Question: What is the name of **RAID** module in a **Linux** kernel?

a) lvm
b) raidsoft
c) md
d) none of the mentioned

Question: What is the minimum number of disks needed for **RAID** level 5?

a) 2
b) 3
c) 4
d) 5

Question: When one disk fails in a **RAID** 5 array with a hot spare, which one of the following statements is true?

a) No IO can continue
b) Reads continue from hot spare, but writes fail
c) Both reads and writes can continue
d) Array must be shut down immediately for replacing the failed disk

---

Question: When one disk fails in a **RAID** 5 array without a hot spare, which one of the following statements is true:

a) No IO can continue
b) Reads continue from hot spare, but writes fail
c) Both reads and writes can continue
d) Array must be shut down immediately for replacing the failed disk

---

Question: Which of the following suffers from write hole problem?

a) RAID 0
b) RAID 1
c) RAID 5
d) RAIDZ

Question: If read and write performance is the only criterion, which would you choose?

a) RAID 0
b) RAID 1

c) RAID 5
d) RAID 50

Question: Which one of the following is an invalid **RAID** level?
a) Raid 10
b) Raid 50
c) Raid 53
d) Raid 80

Question:: Which of the following **RAID** levels guarantees double disk failure protection?
a) Raid 5
b) Raid 6
c) Raid 0+1
d) Raid 0+5

Question:: Which component of an intelligent storage system isolates the host from the mechanical delays associated with rotating disks?

A. Front-end controller
B. Back-end controller
C. Cache
D. Storage network

Question:: Which mode of flushing is activated when the cache reaches 100% of its capacity?

A. Idle
B. High watermark
C. Forced
D. Low watermark

Question::In traditional storage provisioning, which LUN expansion technique provides improved performance?

A. Concatenated metaLUN
B. **Striped metaLUN**
C. Base LUN
D. Component LUN

Question:: Which process provides data access control by restricting host access to specific LUN(s)?

A. LUN masking
B. Zoning
C. Trespassing
D. VSAN

Question:: Which mechanism provides protection to 'uncommitted data in cache' against power failure?

A. Mirroring
B. **Vaulting**
C. Watermarking
D. Tiering

Question:: Consider an application that generates 1200 IOPS at peak workload, with a read/write ratio of 2:1. Calculate disk load at peak activity for RAID 1/0 and RAID 5 configuration.

Solution: RAID Penalty

- **For RAID 1/0, the disk load (read + write)**
  = (1200 x 2/3) + (1200 x (1/3) x 2)
  = 800 + 800
  = 1600 IOPS
- **For RAID 5, the disk load (read + write)**
  = (1200 × 2/3) + (1200 × (1/3) x 4)
  = 800 + 1600
  = 2400 IOPS

Question:: An application generates 400 small random IOPS with a read/write ratio of 3:1. What is the RAID-corrected IOPS on the disk for RAID 5 ?

A. 400
B. 500
C. 700
D. 900

Question:: What is write penalty in a RAID 6 configuration for small random 1/Os?

A. 2
B. 3
C. 4
D. 6

Question:: Which application is most benefited by using RAID 3?

A. Backup
B. OLTP
C. e-commerce
D. email

Question::What is the stripe size of a five disk parity RAID 5 set that has a strip size of 64 KB?

A. 64 K
B. 128 KB
C. 256 KB
D. 320 KB

Question:: Which is a benefit of computer virtualization?

A. Enables compute memory swapping
B. Improves compute utilization
C. Isolates compute memory from the applications
D. Isolates compute OS from the applications

Question:: What best describes virtual machines (VMs)?

A. All VMs on a physical server must run same OS
B. VM files are deleted when VM is powered off
C. VMs are discrete sets of files
D. All VMs share available resources equally

Question:: What is concatenation?

A. **Grouping multiple physical drives into a logical drive**
B. Dividing a physical drive into multiple logical drives
C. Process of writing disk metadata on a logical drive
D. Adding more capacity to a physical drive through defragmentation

---

Question:: Which factors contribute to the overall service time of a mechanical disk?

A. Disk buffer time, full stroke, and rotational latency
B. Internal transfer rate, external transfer rate, and buffer time
C. Full stroke, average seek time, and track-to-track seek time
D. **Average seek time, rotational latency, and data transfer rate**

Question:: Which is a challenge of the DAS environment?

A. Low performance
B. **Limited scalability**
C. Deployment complexity
D. High initial investment

Question:: Which is an example of structured data?

A. Image
B. PDF document
C. **Database**
D. Web page

Question: Which is true about big data?

A. Includes only unstructured data
B. Includes data from a single source
C. Captured efficiently using traditional software tools
D. **Data size is beyond the capability of traditional software to process**

Question: Which is a feature of information-centric architecture?

A. Storage is internal to the servers
B. Prevents sharing of storage among servers
C. Consists of server, network, and storage in a single system
D. **Storage is managed centrally and independent of servers**

Question: What accurately describes virtualization?

A. Provides on-demand, metered services
B. **Abstracts physical resources into logical resources**
C. Pools logical resources to provide data integrity
D. Enables decentralized management across data centers

Question: Which requirement refers to the ability of a storage solution to grow with the business?

A. Availability
B. Manageability
C. Integrity
D. **Scalability**

Question: What enables a logical unit to be presented to a host with more capacity than is physically allocated to it on the storage array?

A. Server Virtualization
B. Thin Provisioning
C. Storage Virtualization
D. File Virtualization

Question: What is the main difference between a native and bridged iSCSI HBA?

A. Native iSCSI includes FC components
B. Bridged iSCSI uses an iSCSI Gateway
C. Bridged iSCSI uses IP protocol, whereas native uses SCSI protocol only
D. Native iSCSI uses FC Host Bus Adaptors

Question: What is true for FCIP?

A. FCIP is good for short distances.
B. FCIP combines iSCSI with IP.
C. FCIP creates virtual FC links in different fabrics.
D. FCIP does not use any gateways.

Question: FC implementations can provide data speeds in the range of

A. 3000 Megabytes per second
B. 3000 Bytes per second
C. 3000 Megabits per second
D. 3000 Kilobits per second

Question: Compared with single mode fiber

A. Multi-mode fiber supports higher data rate
B. Multi-mode fiber has lower efficiency
C. Multi-mode supports longer distances
D. All of the above

Question: Single mode fiber refers to

A. The ability of the fiber filament to support multiple protocols
B. The ability of the fiber to support multiple rays of light
C. The ability of the fiber to support multiple data rates
D. None of the above

Question: When using a fiber channel arbitrated loop

A. The fiber channel bandwidth per user is smaller compared with an FC switch
B. The fiber channel bandwidth per user is higher compared with an FC switch
C. The fiber channel bandwidth per user is equal to that of an FC switch
D. None of the above

Question: In a Fiber Channel network the F_port is used to

A. Connect a storage unit to an FC switch
B. Connect an FC switch to another FC switch
C. Connect an FC switch to an FC hub

D. Connect a storage unit to a server

Question: The FC1 layer is responsible for

A. Data coding and encoding
B. Data routing
C. Data cables and connection
D. None of the above

Question: Using the fiber channel WWN

A. the switch location can be identified
B. the number of ports on the switch can be identified
C. the status of switch can be identified
D. the switch manufacturing company can be identified

Question: Process login (PRLI)

A. allow an N_port to returns an Acc to the initiator N_Port
B. allows a node to obtain an FC address from the switch
C. is performed between two N_ports
D. allows the N_Ports to exchange service parameters relevant to the session

---

Question: Vault drivers are used to

A. Store highly secured data
B. For storing the contents of the cache during power outage
C. To store copy of meta data
D. None of the above

Question: NFS version 4

A. Is based on stateless protocol design
B. Offers enhanced security
C. Provides parallel NFS
D. None of the above

Question: A unified NAS consolidates

A. NAS based and SAN based data access
B. A NAS head and a NAS server
C. A fiber channel NAS with CFI
D. None of the above

Question: ISCSI is widely used because

A. It relatively inexpensive
B. It is easy to implement
C. Both a and b
D. None of a and b

Question: A NAS gateway implementation

A. Uses internal storage
B. Uses external storage
C. Doesn't use storage

D. Doesn't have NAS head

Question: In native iSCSI connectivity

A. **Each device needs to be configures with an IP**
B. Each device needs to be connected to a gateway
C. Each device needs to be connected to SAN first
D. Each device should be connected to fiber channel network

Question: To use FCIP to connect FC host to a FC storage array over an IP network you need

A. At least one bridge
B. At least two bridges
C. At least three bridges
D. At least four bridges

Question: To generate iSCSI qualified Names

A. **An organization must own a registered domain name**
B. An organization must own unique SCSI address
C. An organization must own WWN address
D. None of the above

Question: Two primary protocols that leverage IP as the transport mechanism are

- FTP and HTTP
- **iSCSI and FCIP**
- SAN and NAS
- None of the above

Question: When using the sendTargets discover

- The initiators can automatically register themselves with the iSNS server
- **The initiators respond with the names and addresses of the targets available to**
- The initiator is manually configured with the target's network portal
- The target is manually configures with the initiators network portal

---

Question: NFS version 3

- Uses UDP only
- Uses TCP only
- **Uses TCP and UDP**
- Doesn't use UDP or TCP

Question: iSCSI works on the

- Application layer
- **Transport layer**
- Session layer
- Physical layer

Question:A customer wants to deploy a business continuity solution between two data centers which are500 Km apart. Which solution will provide connectivity between the two FC-SANs in the most cost effective way?

- IP switches communicating over TCP
- Native iSCSI communicating over TCP/IP

- FC switches communicating over FCP
- **FCIP gateways communicating over TCP/IP**

Question: Which protocol is used by a host to access block level storage over IP?
A. iSCSI

Question: Which statement applies to NAS?
A. Specialized server for block level data access

Question: What is a concern of NAS management?
A. SCSI over IP

Question: Why would you implement an iSCSI solution?
B. Leverages existing IP network infrastructure

Question: What is a valid member of a zone in a SAN?

- The IP address of a host
- A Domain ID
- The serial number of a storage array
- **A HBA WWPN**

Question: What is a disadvantage of multi-mode fiber relative to single-mode fiber?
A. Limited distances

Question: What is the protocol used by a Windows client to communicate with a NAS device?
A. CIFS

Question: What is a logical component of a storage environment?
C. Filesystem

Question: Which statement is true about iSCSI names?

- Only a target requires aniSCSI name
- iSCSI name is derived from the IP address
- **Every initiator or target has a uniqueiSCSI name**
- Only an initiator requires aniSCSI name

Question: What is a benefit of a NAS solution?

- Allows the customer to directly connect to the storage array
- **Centralizes storage and simplifies the management**
- Improves network bandwidth
- Allows hosts to access data directly from storage array

Question: How are nodes typically connected in a two tier Core Edge fabric?
Host on Edge, storage on Core

Question: What does a Domain ID identify?
A switch within a fabric

Question: A company has two SAN islands approximately one mile apart. The company wants to create a single fabric over its public WAN connection. Which protocol is recommended to connect the sites?

- 1FCP
- iSCSI
- **FCIP**
- FCP

Question: What distinguishes a FC director from a FC switch?

A. Greater connectivity capacity

Question: What is the advantage of single-mode fiber over multi-mode fiber?

A. Enables optical connections over longer distances

Question: What best describes a Fiber Channel SAN?

Dedicated storage network

Question: Which layer of the protocol stack re-transmits dropped packets in iSCSI?

- IP
- SCSI
- UDP
- **TCP**

Question: What is a reason for implementing single HBA zoning?

- Increase reset time for any change made in the state of the fabric
- Reduce the number of zones required in the SAN
- **Decrease reset time for any change made in the state of the fabric**

Question: What is a purpose of iSNS?

A. To assist with discovery of iSCSI targets

Question: What is a reason for implementing single HBA zoning?

Minimizes impact to nodes after an RSCN

Question: Which SAN management feature prevents unauthorized or accidental access to volumes?

- LUN Masking
- Secure Fabric OS
- Fibre Channel Persistent ID
- **Single HBA Zoning**

Question: In the context of protecting the FC storage infrastructure, what is an example of infrastructure integrity?

D. Preventing node login to the fabric without proper authorization

---

Question: What is the requirement for native iSCSI implementation?

- Host withTo card and array with iSCSI port
- **Host with softwareiSCSI initiator and array with FCIP port**

Question: How many devices can be connected to a storage array in a point-to-point configuration?

A. 2

Question: How many domain ids are reserved in fiber channel address space?

A. 17 (256-239)

Question: Which cable type provides minimum signal attenuation over long distance?

- Twisted-pair copper
- Coaxial copper
- **Single-mode optical**
- Multimode optical

Question: What is an F_Port in FC SAN?

- Switch port that connects an E_Port
- **Switch port that connects an N_Port**
- Node port that connects an N_Port
- Node port that connects an E_Port

Question: Which type of fabric login enables the exchange of upper layer protocol-related parameters between N_Ports?

- Fabric login
- Port login
- **Process login**
- ULP login

Question: Which is a benefit of zoning in FC SAN?

- Isolates fabric services
- **Restricts RSCN traffic**
- Enables online volume expansion
- Provides non-disruptive data migration

Question: Which is a benefit of VSAN?

- Eliminates need for fabric login process
- Provides higher network bandwidth
- **Improves security by isolating traffic between VSANs**
- Enables VSANs to share fabric zoning service

Question: Which iSCSI host connectivity option offloads both iSCSI and TCP/IP processing from the host CPU?

- Standard NIC with isCSI initiator software
- TOE NIC
- **ISCSI HBA**
- CNA

Question: Which type of iSCSI name requires a registered domain name to generate unique isCSI identifier?

- eui
- **iqn**
- WWN
- MAC

Question: Which protocol encapsulates FC frames onto IP packet?

- FCOE
- iSCSI
- **FCIP**
- CIFS

Question: Which component of a NAS head translates file-level requests into block-storage requests?

- Front-end ports
- **Optimized operating system**
- CIFS and NFS
- Network Interface Card

Question: Which is a feature of scale-out NAS?

- Uses general purpose operating system for file serving
- Creates multiple file systems on each node in the cluster
- Uses external and independently-managed nodes
- **Enables pooling of nodes that work as a single NAS device**

Question: Which is a feature of gateway NAS?

- Uses dedicated storage for each NAS head
- **NAS head and storage are managed independently**
- Creates a single file system that runs on all NAS heads
- Provides connectivity to iSCSI and FC hosts

Question: Which NAS implementation consolidates file-based and block-based access on a single storage platform?

- Scale-out
- Gateway
- **Unified**
- Both gateway and scale-out

Question: Which is a benefit of file-level virtualization?

- Enables users to use physical path, rather than logical path, to access files
- Translates file-level request into block-storage request non disruptively
- Consolidates NAS-based and SAN-based access on a single storage platform
- **Eliminates dependency between data accessed at file level and file location**

Question: An array of disks is more likely to fail compared to a single disk. How is it that RAID arrays still manage to provide more data protection compared to a single disk?

a. Using either mirroring or striping
b. Using either mirroring or parity
c. Using better quality disks
d. Using dedicated hardware

Question: The mean time to failure of a single disk is 200000 hrs. If an array of disks is using 5 such disks, the mttf of the array is

a. 200000 hrs
b. 40000 hrs
c. 1000000 hrs
d. Can not be determined

Question: Can you help decide on the RAID level to use – we are a media house and we use lot of graphics/video applications – we need large throughputs for videos to get played without any jitter and since we are in the publishing business we can't afford downtimes.
Even if there is any downtime we would like our data to be quickly reconstructed and enable us to continue without work in less time
a. Raid 5
b. Raid 10
c. Raid 6
d. Raid 01
e. Raid 0

Question: Which of the following raid levels provides maximum usable disk space?
a. RAID 1
b. RAID 0
c. RAID 5
d. RAID 6

Question: What is the minimum number of disks required for RAID1?
a. 1
b. 2
c. 4
d. 5

Question: Which of the following combinations can support RAID 05?
a. 2 sets with 3 disks each
b. 3 sets with 2 disks each
c. 4 sets with 3 disks each
d. 4 sets with 1 disk each

Question: What is the unique characteristic of RAID 6 (Choose one)?
a. Distributed Parity
b. Striping
c. Two independent distributed parity
d. Mirroring

Question: Which one of these is characteristic of RAID 5?
a. Distributed parity
b. No Parity
c. All parity in a single disk
d. Double Parity

Question: Which two RAID types use parity for data protection?
a. RAID 1

b. RAID 4
c. RAID 1+ 0
d. RAID 5

Question: Which two are advantages of hardware RAID controllers?
a. Volume management is performed by the server
b. Volume management is performed by controller card
c. Dedicated cache memory increases server write performance
d. Parity calculation by the server and cache memory in the RAID controller increases read and write performance.

Question: Pick the false statement
a. RAID Level 1 provides disk mirroring
b. RAID Level 2 provides bit level striping with Hamming code ECC
c. RAID Level 4 provides block level striping
d. RAID Level 5 provides block level striping and error correction information

---

Question: Which of the following provides byte level striping?
a. RAID 6
b. RAID 4
c. RAID 2
d. RAID 5

Question: State true or false
a. Nested RAID provides better performance characteristics than the RAID levels that comprise them (true/false) – **True**
b. RAID Level 6 has dual parity where as RAID Level 5 has single parity and both does block level striping(true/false) – **False**
c. RAID Level 7 does Asynchronous cached striping with dedicated parity(true/false) – **True**

Question: Which of the following is false about tape devices?
a. A tape drive is a data storage device that reads and writes data stored on a Magnetic tape
b. Tape drives are used for archival storage of data
c. Tape media has low unit cost and long archival stability
d. Tape drives allow random access of data

Question: Select the correct option regarding tape drive
a. Tape drives have fast average seek times
b. Tape drives can stream data at a very fast rate
c. Archival life of data stored on tape is around 5 years
d. Tape drives can be used to store data on optical disks

Question: A tape library does not contain
a. Tape Drive
b. Bar Code Reader
c. Robotic Arm
d. RAID Array

Question: Which of the following is not true about a tape silo?

a. Libraries provide large storage capacity at a very cheap rate

b. They have slow access time

c. Tape libraries are primarily used for backups and as the final stage of digital archiving

---

Question: For long term storage (archival) of data which of the following storage devices is generally used

a. Hard Disk

b. CD Rom

c. Floppy Disk

d. Tape Cartridges

Question: Which of the following Company manufactures Tape Library?

a. Quantum

b. Brocade

c. NetApps

d. Cisco

Question: Which of the following Company manufactures RAID devices?

a. Qlogic

b. LSI

c. Falconstor

d. Quantum

Question: Which data center characteristic ensures that adequate resources are provided to efficiently store and process data?

A. Capacity

B. Availability

C. Scalability

D. Performance

Question: Which data center management process involves collating and presenting the utilization of resources? A. Reporting

B. Provisioning

C. Planning

D. Maintenance

Question: Which type of digital data has no defined format but has a self-describing structure that enables its analysis?

A. Semi-structured data

B. Structured data

C. Metadata

D. Quasi-structured data

Question: Which type of digital data consists of textual data with inconsistent formats but can be formatted with the use of software tools?

A. Quasi-structured data

B. Semi-structured data

C. Metadata

D. Unstructured data

Question: What are the two [2] primary reasons why businesses collect and store data over long periods of time?

a. b. Availability of high capacity storage systems

Availability of high performance servers

c. Potential for new business initiatives

d. Regulations and compliance

e. Availability of low cost storage system

Question: What are two [2] advantages of centralized networked storage?

a. Consolidated data storage and retrieval

b. Islands of data

c. Improved storage performance

d. Multiple dedicated storage arrays

e. Sharing storage resources and data between different departments in an organization

---

Question: Which storage systems key requirement refers to the ability of the storage solution to grow with the business?

a. Availability

b. Capacity

c. Data Integrity

d. Scalability

Question: What is Read Hit Ratio?

a. Number of misses with respect to the total read request

b. Number of hits with respect to the total write request

c. Number of hits with respect to the total read request

d. Number of misses with respect to the total write request

Question: Which statement best describes little's law?

a.Arrival rate is directly proportional to response time

B. Arrival rate is inversely proportional to total number of request in the system

C. Response time is inversely proportional to total number of request in the system

d. Arrival rate is inversely proportional to response time

Question: What two [2] factors or conditions determine the storage capacity of a Head Disk Assembly (HDA)?

a. Amount of data that can be stored on each platter

b. c. Cache size of the platters

C. Connectivity of platters to the HDA

d. Number of platters

e. Size of HAD

Question: What application I/O characteristic is most suitable for use of RAID 3?

a. Small random reads

b. Large sequential reads

c. Large random writes

d. Small sequential writes

Question: What is the primary benefit of data striping?

a. Independent disks allow multiple reads/writes simultaneously

b. Improves security by creating multiple identical copies of data

C. More efficient allocation of sequential reading/writing

D. More efficient allocation of random reading/writing

Question: Which term is used to describe an idle component in a RAID array that becomes a temporary replacement for a failed component?

a. Hot disk

b. Hot spare

c. Hot swap

d. RAID controller

Question: What is the role of the front-end in an intelligent storage system?

a. Improves system performance by isolating disks from the mechanical delays associated with physical disks

b. Provides temporary storage of data before committing to disk

c. Provides connectivity from the host to the storage array

d. Provides communication with the disks for read/write operations

---

Question: Which component of an intelligent storage system improves performance of reads and writes?

a. Back-end

b. Cache

c. Host

d. Physical Disks

Question: Which activity represents ILM strategy?

a. Automate backup and recovery process

b. Implement storage tiering

c. Migrate from DAS to SAN

d. Implement end-to-end storage management

Question: What is the purpose of cache vaulting?

a. Protect uncommitted data due to cache failure

b. Backup data at offsite location

c. Protect uncommitted data due to power failure

d. Provides better cache management

Question: Which key parameter decides the number of disks in a RAID group?

a. Capacity required

b. RAID type

c. Number of IOPS

d. Stripe depth

Question:A host generates 1000 IOPS of which 60% are writes. In a RAID 5 configuration, what is the

number of IOPS on the disk?
a. 600 IOPS
b. 1600 IOPS
c. 2200 IOPS
d. 2800 IOPS

Question: Process of storing object in an OSD system in shown above which step is missing

1. The OSD node generates die file ID using a specialised algorithm
2. The OSD node receives the object ID
3. **The OSD node generates the object ID using a specialized algorithm**
4. None of the above

Question: Benefits of SAN include

1. **Centralized control. Policy-based automation Simplified management**
2. Distributed control. Policy-based automation Simplified management
3. Centralized control. Policy-based automation hierarchical management
4. Centralized control. Acceptance-based automation Simplified management

---

Question: storing data in the form of objects on flat address space based on its content and other attributes rather that the name and the location is called

1. **Object defined storage**
2. Space less defined storage
3. Unified defined storage
4. Software defined storage

Question: Multi-tenancy in OSD means

1. policy-based information management capabilities combine to intelligently (automate) drive data placement, data protection,
2. **Enables multiple applications to be securely served from the same infrastructure**
3. Enables single applications to be securely served from the same infrastructure
4. None of the above

Question: to have an integrated storage solution that supports block, file, and object access is called

1. Multi storage
2. Combine storage
3. **Unified storage**
4. Integrated storage

Question: SAN stands for

1. **Storage area network**
2. Storage advanced network
3. Single area network
4. Source area network

Question: SAN is implemented over

1. Fiber channel
2. TCP

3. Ethernet
4. **Both 1 and 3**

Question: FC SAN Interconnectivity options include

1. Point to many
2. Arabiated link
3. **Point to point**
4. None of the above

Question: A customer has a database application that performs 5000 IOPS with segment size 1 KB. This application is a time critical application and needs storage capacity of 100 TB.
The available hard disk in the market costs 200 US $ and has the below specifications:

- Full stroke seek time is 51 ms
- RPM is 15k
- Disk Data rate is 15 MBps
- Capacity is 250 GB

The customer has decided to apply RAID 5 in the storage server, but has budget limit of 90,000 US $. Find the minimum number of hard disks that can share the same parity in this RAID 5 implementation. (5 points)

Solution:
No. of hard disks "from Capacity"= 100T/0.25T = 400 HDs
HD service time= Average Seek time + Average rotation time+ transfer time
= 1/3 * Full stroke + 0.5 * 1/ (RPM/60) + segment size/ transfer rate
= (1/3)*(51ms) + 0.5* (1/ (15*103/60)) + 103/ (15*106) = 19 ms
IOPS per HD = 52.63

Total No. of IOPS= 5000*3/5 + 4*5000*2/5= 11000
No. of hard disks "from IOPS»=11000/52.63= 209
So, the required number of HDs = 400
Total number of HDs after RAID 5 implementation = 400*(N+1)/N ; where N is the number of HDs share the same parity.
From the budget limit, Max. number of HDs= 90,000/200 = 450 HDs.
So 450 = 400*(N+1)/N → N=8

Question: Which of the following is correct about hardware RAID controllers?
a) Volume management is performed by the server
b) Volume management is performed by controller card
c) Dedicated cache memory decreases server write performance
d) Parity calculation by the server and cache memory in the RAID controller increases read and write performance

---

Question: Which RAID type doesn't use parity for data protection?
a) RAID 1
b) RAID 4
c) RAID 6
d) RAID 5

Question: Which of the below is component of SAN
a) Ethernet switch
b) Fibre Switch
c) Ethernet Card
d) None of the mentioned

Question: What components on SAN can be virtualized
a) Disk
b) Tape Drive
c) Tape Library
d) All of the mentioned

Question: C-H-S is a terminology related to
a) Logical Disks
b) Physical Disk
c) Physical Volume
d) None of the mentioned

Question: Which device implements Tape-Over-Tape Virtualization
a) Falconstor VTL
b) VSM Open
c) SN6000
d) None of the mentioned

Question: What is one significant advantage of Virtualization in datacenters
a) Can manage the Data center with Zero Manpower
b) Mirroring for Disaster Recovery
c) Eliminate the need of physical devices for the Data center
d) None of the mentioned

Question: Powerpath is the MPIO software of
a) Microsoft
b) Veritas
c) EMC
d) None of the mentioned

Question: Which is the correct statement
a) Virtualization cloud presents itself as host to the host
b) Virtualization cloud presents itself as targets to the host
c) Virtualization cloud can be set up to present itself has either host or target to the host
d) None of the mentioned

Question: Decoupling the Host and real targets in a datacenter is a bliss because it
a) Allows dynamic configuration of targets without host knowing about it
b) Allows dynamic of configuration of hosts without targets knowing about it
c) Allows static configuration of targets with host knowing about of it
d) None of the mentioned

Question: A SAN network can have
a) 1 FC switch
b) 2 FC switch
c) There can be many FC switches
d) No FC switch

Question: This is a repository for the storage, management, and dissemination of data in which the mechanical,
lighting, electrical and computer systems are designed for maximum energy efficiency and minimum environmental impact.
a) Storage lab
b) Data Center
c) Data warehouse
d) Fabric

Question: This is the process of assigning storage, usually in the form of server disk drive space, in order to optimize the performance of a storage area network.
a) Storage Provisioning
b) Data mining
c) Storage assignment
d) Data Warehousing

Question: Simply stated, these are large boxes that hold lots of hard disks.
a) Host
b) Tape library
c) Switch
d) Disk Array

Question: This consists of the precautions taken so that the effects of a disaster will be minimized.
a) Data retrieval
b) Disaster recovery
c) Archive
d) Replication

Question: This is the practice of collecting computer files that have been packaged together for backup, to transport to some other location, for saving away from the computer so that more hard disks can be made available, or for some other purpose.
a) Backup
b) Archive
c) Migration
d) Compression

Question: What is Cloud Computing?
a) Cloud Computing means providing services like storage, servers, database, networking, etc
b) Cloud Computing means storing data in a database
c) Cloud Computing is a tool used to create an application
d) None of the mentioned

Question: Which of the following is not a type of cloud server?
a) Public Cloud Servers
b) Private Cloud Servers
c) Dedicated Cloud Servers
d) Merged Cloud Servers

Question: Which of the following are the features of cloud computing?
a) Security
b) Availability
c) Large Network Access
d) All of the mentioned

Question: Which of the following is a type of cloud computing service?
a) Service-as-a-Software (SaaS)
b) Software-and-a-Server (SaaS)
c) Software-as-a-Service (SaaS)
d) Software-as-a-Server (SaaS)

Question: Which of the following is the application of cloud computing?
a) Adobe
b) Paypal
c) Google G Suite
d) All of the above

Question: Which of the following is an example of the cloud?
a) Amazon Web Services (AWS)
b) Dropbox
c) Cisco WebEx
d) All of the above

Question: Which of the following is an example of a PaaS cloud service?
a) Heroku
b) AWS Elastic Beanstalk
c) Windows Azure
d) All of the above

Question: Which of the following is an example of an IaaS Cloud service?
a) DigitalOcean
b) Linode
c) Rackspace
d) All of the above

Question: Which of the following is the correct statement about cloud computing?
a) Cloud computing abstracts systems by pooling and sharing resources
b) Cloud computing is nothing more than the Internet
c) The use of the word “cloud” makes reference to the two essential concepts
d) All of the mentioned

Question: Point out the wrong statement.
a) Azure enables .NET Framework applications to run over the Internet
b) Cloud Computing has two distinct sets of models
c) Amazon has built a worldwide network of datacenters to service its search engine
d) None of the mentioned

Question: Which of the following is the correct statement about cloud types?
a) Cloud Square Model is meant to show is that the traditional notion of a network boundary being the network's firewall no longer applies in cloud computing
b) A deployment model defines the purpose of the cloud and the nature of how the cloud is located
c) Service model defines the purpose of the cloud and the nature of how the cloud is located
d) All of the mentioned

Question: Which architectural layer is used as a backend in cloud computing?
a) cloud
b) soft
c) client
d) all of the mentioned

Question: Which of the following architectural standards is working with cloud computing industry?
a) Web-application frameworks
b) Service-oriented architecture
c) Standardized Web services
d) All of the mentioned

Question: Which of the following is the correct statement?
a) Cloud computing presents new opportunities to users and developers
b) Service Level Agreements (SLAs) is small aspect of cloud computing
c) Cloud computing does not have impact on software licensing
d) All of the mentioned

Question: Which of the following is the wrong statement about cloud computing?
a) Private cloud doesn't employ the same level of virtualization
b) Data center operates under average loads
c) Private cloud doesn't pooling of resources that a cloud computing provider can achieve
d) Abstraction enables the key benefit of cloud computing: shared, ubiquitous access

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Question: Identify the wrong statement about cloud computing.
a) Virtualization assigns a logical name for a physical resource and then provides a pointer to that physical resource when a request is made
b) Virtual appliances are becoming a very important standard cloud computing deployment object
c) Cloud computing requires some standard protocols
d) None of the mentioned

Question: Identify the correct statement about cloud computing.

a) Cloud computing relies on a set of protocols needed to manage interprocess communications
b) Platforms are used to create more complex software
c) Cloud architecture can couple software running on virtualized hardware in multiple locations to provide an on-demand service
d) All of the mentioned

Question: Which of the following is required by Cloud Computing?
a) That the identity be authenticated
b) That the authentication be portable
c) That you establish an identity
d) All of the mentioned

Question: Cloud computing is a concept that involves pooling physical resources and offering them as which sort of resource?
a) cloud
b) real
c) virtual
d) none of the mentioned

Question: Which of the following is the Cloud Platform provided by Amazon?
a) AWS
b) Cloudera
c) Azure
d) All of the mentioned

Question: Which of the following is the most essential element in cloud computing by CSA?
a) Virtualization
b) Multi-tenancy
c) Identity and access management
d) All of the mentioned

Question: Which of the following model consists of the service that you can access on a cloud computing platform?
a) Deployment
b) Service
c) Application
d) None of the mentioned

Question: Which of the following is the most important area of concern in cloud computing?
a) Scalability
b) Storage
c) Security
d) All of the mentioned

Question: Which of the following is the most refined and restrictive cloud service model?
a) PaaS
b) IaaS

c) SaaS
d) CaaS

Question: Which of the following is not a property of cloud computing?
a) virtualization
b) composability
c) scalability
d) all of the mentioned

Question: How many phases are there in Cloud Computing Planning?
a) 1
b) 5
c) 3
d) 6

Question: Which of the following is an example of a SaaS cloud service?
a) Google Workspace
b) Dropbox
c) Salesforce
d) All of the above

Question: Which is the most essential concept related to Cloud computing?
a) Abstraction
b) Reliability
c) Productivity
d) All of the mentioned

Question: In which of the following service models the hardware is virtualized in the cloud?
a) NaaS
b) PaaS
c) CaaS
d) IaaS

Question: Which of the following is the Virtual machine conversion cloud?
a) Amazon CloudWatch
b) AbiCloud
c) BMC Cloud Computing Initiative
d) None of the mentioned

Question: An application that provides for transaction overflow in a reservation system is an example of _____
a) cloud bursting
b) cloud provisioning
c) cloud servicing
d) all of the mentioned

Question: Which of the following is owned by an organization selling cloud services?
a) Public

b) Private
c) Community
d) Hybrid

Question: Point out the wrong statement.
a) Everything from the application down to the infrastructure is the vendor's responsibility
b) In the deployment model, different cloud types are an expression of the manner in which infrastructure is deployed
c) AaaS provides virtual machines, operating systems, applications, services, development frameworks, transactions, and control structures
d) All of the mentioned

Question: _____ provides virtual machines, virtual storage, virtual infrastructure, and other hardware assets.
a) IaaS
b) SaaS
c) PaaS
d) All of the mentioned

Question: Which of the following provides development frameworks and control structures?
a) IaaS
b) SaaS
c) PaaS
d) All of the mentioned

Question: Point out the wrong statement.
a) A PaaS service adds integration features, middleware, and other orchestration and choreography services to the IaaS model
b) XaaS or 'anything as a service' is the delivery of IT as a Service through hybrid Cloud computing
c) Monitoring as a Service (MaaS) is at present still an emerging piece of the Cloud jigsaw
d) None of the mentioned

Question: _____ is a complete operating environment with applications, management, and user interface.
a) IaaS
b) SaaS
c) PaaS
d) All of the mentioned

Question: How many types of service model are mainly present in Cloud?
a) 1
b) 2
c) 3
d) 4

Question: The three different service models are together known as the _____ model of cloud computing.
a) SPI
b) SIP

c) CPI
d) All of the mentioned

Question: CaaS stands for _____ as service.
a) Compliance
b) Computer
c) Community
d) Communication

Question: Which of the following is the IaaS service provider?
a) EC2
b) EC1
c) EC10
d) Hybrid

Question: Point out the wrong statement.
a) Public cloud may be managed by the constituent organization(s) or by a third party
b) A community cloud may be managed by the constituent organization(s) or by a third party
c) Private clouds may be either on- or off-premises
d) None of the mentioned

Question: Which of the following is related to the service provided by Cloud?
a) Sourcing
b) Ownership
c) Reliability
d) AaaS

Question: _____dimension corresponds to two different states in the eight possible cloud forms.
a) Physical location of data
b) Ownership
c) Security boundary
d) None of the mentioned

Question: The _____cloud infrastructure is operated for the exclusive use of an organization.
a) Public
b) Private
c) Community
d) All of the mentioned

Question: _____cloud is one where the cloud has been organized to serve a common function or purpose.
a) Public
b) Private
c) Community
d) All of the mentioned

Question:A hybrid cloud combines multiple clouds where those clouds retain their unique identities but are bound together as a unit.
a) Public
b) Private
c) Community
d) Hybrid

Question: Google Docs is a type of cloud computing.
A. True
B. False

Question: Which of these techniques is vital for creating cloud-computing centers?
A. Virtualization

Question: This is a software distribution model in which applications are hosted by a vendor or service provider and made available to customers over a network, typically the Internet.
C. Software as a Service (SaaS)

Question: Virtual Machine Ware (VMware) is an example of
A. Infrastructure Service

Question: Cloud Service consists of
A. Platform, Software, Infrastructure

Question:A file in HDFS that is smaller than a single block size
D. occupies only the size it needs and not thefull block

Question: For the frequently accessed HDFS files the blocks are cached in
A. the memory of thedata node

Question: Which among the following is the duties of the Data Nodes
D. perform read-write operation as per request for the clients

---

Question: In HDFS the files cannot be
C. executed

Question: Which of the following storage devices exposes its storage to clients as Raw storage that can be partitioned to create volumes ?
A. block

Question: The process of duplicating every disk is called_____
c) Mirroring

Question: Data striping by splitting bits across multiple disks is called as _____
c) Bit level striping

Question: How many levels of RAID exist?
c) 7

Question: RAID level 0 refers to?

a) Disk arrays with striping at the level of blocks

Question: RAID level 1 refers to?

b) Disk mirroring with block striping

Question: RAID level 5 refers to?

d) Block interleaved distributed parity

Question: _____ is popular for applications such as storage of log files in a database system since it offers the best write performance.

a) RAID level 1

Question: _____ which increases the number of I/O operations needed to write a single logical block, pays a significant time penalty in terms of write performance.

a) RAID level 1

Question:RAID level 3 supports a lower number of I/Os per second, because _____

a) Every disk has to participate in every I/O request2.

Question:RAID level _____ is also known as block interleaved parity organisation and uses block level striping and keeps a parity block on a separate disk.

d) 4

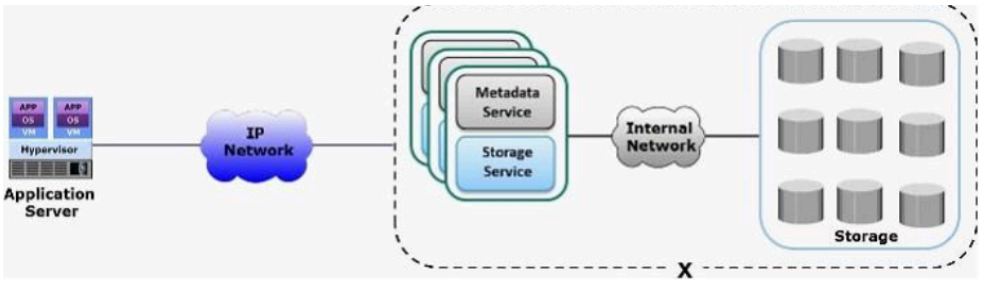
Question:A performance problem with _____ is the expense of computing and writing parity.

b) parity based RAID levels

Question:RAID level ____ spreads parity and data among all N+1 disks rather than storing data in N disks and parity in 1

c) 5

Question: Which type of storage device does the "X" represent in the exhibit?



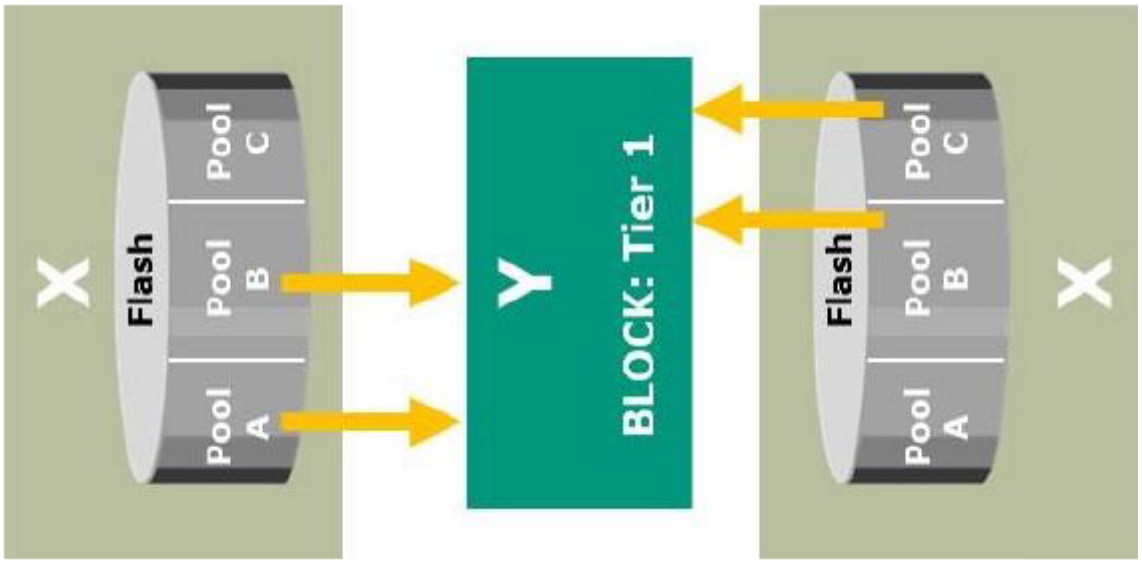
A. Object-based

B. Block-based

C. File-based

D. Unified-based

Question: The exhibit illustrates a software-defined storage environment. What do the labels “X” and “Y” represent in the exhibit?



A. X is a virtual storage system and Y is a virtual storage pool
B. X is a virtual storage pool and Y is a virtual storage system
C. X is a physical storage pool and Y is a virtual storage system
D. X is a virtual storage pool and Y is a physical storage system

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Question: A start-up company with a limited budget is planning to adopt a hybrid cloud solution for their operations. The company has business-critical applications that serve customers and have strict service levels. In addition, the company has less critical applications such as
backup and archive.
Which hybrid cloud strategy should be recommended to the company?
A. Deploy the business-critical applications on an externally-hosted private cloud. Deploy the less critical applications on a public cloud.
B. Deploy the business-critical applications on an on-premise private cloud. Deploy the less critical applications on an externally-hosted private cloud.
C. Deploy the business-critical applications on an on-premise private cloud. Deploy the less critical applications on a public cloud.
D. Deploy the business-critical applications on an externally-hosted private cloud. Deploy the less critical applications on an on-premise private cloud.

Question: What is a benefit of using converged infrastructure for building a data center?
A. Reduces the time to acquire and deploy the infrastructure
B. Provides the flexibility to purchase individual IT components
C. Integrates existing IT components into the infrastructure
D. Offers the flexibility to change vendors and prevent vendor lock-in

---

Question: What is a benefit of a storage area network (SAN)?
A. Enables compute systems across locations to access shared storage
B. Ensures secure data transmission using only channel technologies
C. Ensures high response times without the protocol overhead
D. Enables consolidation of control and data planes in a single system

---

Question: Which data center management process involves collating and presenting the utilization of resources?
A. Reporting

B. Provisioning
C. Planning
D. Maintenance

Question: An organization plans to deploy a new application in their environment. The new application requires 2 TB of storage space. During peak workloads, the application is expected to generate 9800 IOPS with a typical I/O size of 4 KB. The capacity of each
the available disk drive is 250 GB. The maximum number of IOPS a drive can perform at with 70 percent utilization is 90 IOPS.
What is the minimum number of disk drives needed to meet the application's capacity and performance requirements given a RAID 0 configuration?
A. 8
B. 70
C. 110
D. 140

Question: Which type of LUN provides full control for precise data placement and allows an administrator to create LUNs on different RAID sets if there is workload contention?
A. Traditional
B. Thin
C. Journal
D. Bitmap

Question: What is signified by the "variety" characteristic of big data?
A. Data is generated in different formats by numerous sources
B. Meaning of the data changes constantly
C. Varying data quality affects its reliability and accuracy
D. Data change rate affects its timely analysis

Question: Which type of storage system is built specifically for storing fixed content?
A. Content addressed storage
B. Block-based storage
C. File-based storage
D. Unified storage

Question: Which capability does big data analytics provide?
A. Enabling data-driven decisions from large volumes of data
B. Storing large volumes of data without affecting availability
C. Reducing the data storage capacity requirement
D. Performing analytics using traditional tools

Question Which VMware product provides a network virtualization platform for a software-defined data center?
A. NSX
B. ESXi
C. Horizon
D. AirWatch

Question Which capability is provided by a software-defined storage controller?

A. Ability to pool and abstract physical storage and present it as an open storage platform

B. Ability for multiple operating systems to run concurrently on a single physical compute system

C. Mechanisms to automatically create application programming interfaces for centralized management

D. Single self-contained package with integrated hardware and software components

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Question Which data center characteristic ensures that data is stored and retrieved exactly as it was received?

A. Data integrity

B. Security

C. Scalability

D. Manageability

Question: What is a characteristic of a scale-out NAS?

A. File system grows dynamically as nodes are added in the cluster

B. Up to four file systems can be created across the cluster

C. Distinct file systems are created on each node in the cluster

D. Different file systems can be mixed on each node in a single cluster

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Question: What is an accurate statement about full volume replication?

A. Target device is at least as large as the source device

B. Target device requires only a fraction of the space used by the source device

C. Target device is accessible only after the replication session starts

D. Target and source devices should be on different storage arrays

---

Question: What is an example of an active attack on an IT infrastructure?

A. Attempt to modify information for malicious purposes

B. Attempt to copy the data from the storage system

C. Attempt to gain unauthorized access into the system

D. Attempt to gain access to the data in transit

Question: What accurately describes link aggregation in a SAN?

A. Combines two or more parallel interswitch links into a single logical interswitch link

B. Aggregates the data transmission rate of all HBA and storage system port pairs

C. Consolidates compute-to-compute traffic and storage traffic into a single interswitch link

D. Replaces multiple parallel interswitch links by a single physical interswitch link

---

Question: Which solid state drive component performs write coalescing?

A. Controller

B. I/O interface

C. Non-volatile memory

D. Cache

Question: In an object-based storage system, data is divided into 9 data segments and 3 coding segments using the erasure coding technique. What is the maximum number of disk drive failures against which data is protected by this configuration?

A. 3
B. 6
C. 9
D. 12

Question: What is an accurate statement about Platform as a Service?
A. Consumer has control over the deployed applications
B. Consumer has control over the operating system and database management system
C. Consumer has control over the programming languages and tools
D. Consumer has control over the operating system and storage

---

Question: An application generates 8400 small random I/Os at peak workloads with a read/write ratio of 2:1. What is the disk load at peak activity for a RAID 5 configuration?
A. 4800
B. 11200
C. 16800
D. 22400

Question: Which feature of an object-based storage system abstracts and provides a common view of storage independent of location to applications?
A. Global namespace
B. Flexible data access
C. Multi-tenancy
D. Scale-out architecture

Question: What is an accurate statement about three-way NDMP backup in a NAS environment?
A. Backup device can be shared among multiple NAS heads
B. Each NAS head should have a dedicated backup device
C. Backup data traverses at least three NAS heads before reaching the backup device
D. Backup is performed only when three or more NAS heads are present

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Question: What is an accurate statement about a data lake?
A. Stores data as an exact or near-exact copy of the source format
B. Supports query and analysis by classifying and organizing data before storing
C. Stores current and historical data in a structured format on object-based storage
D. Provides a single consistent view of data across an organization for report generation

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Question: Which benefit does the measured service characteristic provide to a cloud service provider?
A. Enables control and optimization of resource use
B. Enables cloud services to communicate with each other
C. Enables availability of specific resources depending on policy
D. Enables the reduction and/or elimination of upfront IT expenditure

---

Question: An application generates 8400 small random I/Os at peak workloads with a read/write ratio of 2:1. What is the disk load at peak activity for a RAID 1/0 configuration?
A. 5600
B. 11200
C. 16800

D. 22400

Question: An application generates 4200 small random I/Os at peak workloads with a read/write ratio of 2:1. What is the disk load at peak activity for a RAID 6 configuration?

A. 2800

B. 5600

C. 8400

D. 11200

Question: Which elements are associated with an object in an object-based storage system?

A. Data, metadata, object ID, and attributes

B. LUN ID, data, metadata, and object ID

C. Metadata, object ID, attributes, and file path

D. Object ID, attributes, LUN ID, and file path

Question: An organization plans to deploy a new application in their environment. The new application requires 4 TB of storage space. During peak workloads, the application is expected to generate 4900 IOPS with a typical I/O size of 8 KB. The capacity of each
the available disk drive is 500 GB. The maximum number of IOPS a drive can perform at with a 70 percent utilization is 110 IOPS.
What is the minimum number of disk drives needed to meet the application's capacity and performance requirements given a RAID 0 configuration?

A. 8

B. 20

C. 30

D. 45

Question: What is an accurate statement about a virtual machine (VM) snapshot?

A. Captures the state and data of a running VM at a specific point-in-time

B. Provides the capability to delete the parent VM once the snapshot is created

C. Runs as an independent copy from the parent VM

D. Provides the capability to restore VM data to any point-in-time

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Question: What is a function of the orchestration layer in a data center infrastructure?

A. Provides workflows for automated execution of management tasks

B. Aggregates physical infrastructure components into resource pools

C. Measures the consumption of IT resources by various services

D. Decouples an operating environment from the underlying hardware

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Question: What maps the logical path of a file system to its physical path in a file-level virtualization implementation?

A. Global namespace

B. Remote procedure call

C. Distributed file system

D. Cache coherency

Question: An organization performs a full backup every Sunday and incremental back ups on the remaining days at 9:00 AM. However, at 10 PM on Friday, a data corruption occurs that

requires a data restoration from the backup copies.
How many backup copies are required to perform this restoration?
A. 2
B. 4
C. 5
D. 6

Question: An organization performs a full backup every Sunday and cumulative back ups on the remaining days. On Thursday, a data corruption occurs that
requires a data restoration from the backup copies.
How many backup copies are required to perform this restoration?
A. 1
B. 2
C. 3
D. 4

Question: What is an accurate statement about a logical volume in a compute system?
A. Can span across multiple physical volumes
B. Can only be created from a single physical volume
C. Must be created across multiple volume groups
D. Allows multiple file systems to run concurrently

Question: What accurately describes the seek time of a disk drive?
A. Time taken to position the read/write head over the correct track
B. Time taken by the platter to position the data under the read/write head
C. Time taken to read a block of data from the platter
D. Time taken by the controller to service an I/O from the application

---

Question: What is an accurate statement about a hardware RAID implementation?
A. Uses a specialized hardware controller installed on a storage array
B. Uses CPU cycles of a compute system to perform RAID calculations
C. Requires a RAID controller and operating system to be compatible
D. Offers a low-cost solution for providing disk protection

Question: What is an accurate statement about a software RAID implementation?
A. Uses CPU cycles of a compute system to perform RAID calculations
B. Supports the upgrade of RAID software on incompatible O/S
C. Uses a specialized hardware controller to perform RAID calculations
D. Supports all RAID levels

Question: What is a function of the metadata manager in a compute-based storage area network (SAN)?
A. Controls load balancing and data migration in a compute-based SAN
B. Provides an interface to the software-defined storage controller
C. Controls all data read and write operations in a compute-based SAN
D. Performs the I/O operations on the local storage of a compute system as requested by a client

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Question:A system has three components and all three components need to be operational for 24 hours; Monday through Friday. During a particular week, failure of component 3 occurred
as follows:

Tuesday = 8 AM to 1 PM
Thursday = 5 PM to 10 PM
Friday = 5 AM to 10 AM
What is the mean time between failures on component 3 for that week?
A. 5 hours
B. 35 hours
C. 40 hours
D. 51 hours

Question: A system has three components and all three components need to be operational for 24 hours; Monday through Friday. During a particular week, failure
of component 1 occurred as follows:
Tuesday = 8 AM to 11 AM
Thursday = 8 PM to 10 PM
Friday = 5 AM to 9 AM
What is the mean time between failures on component 1 for that week?
A. 3 hours
B. 37 hours
C. 40 hours
D. 53 hours

Question: In an object-based storage system, data is divided into 10 data segments and 4 coding segments using the erasure coding technique. What is the maximum number of disk drive failures against which data is protected by this configuration?
A. 4
B. 6
C. 10
D. 14

Question: What provides the ability to automatically relocate extents across physical storage drives over the entire pool when new drives are added to the pool?
A. Storage pool rebalancing
B. Extent relocation
C. Storage pool tiering
D. Automated extent tiering

Question: What is a benefit of link aggregation in a SAN?
A. Provides a higher throughput than a single interswitch link can provide
B. Distributes the network traffic across all virtual SANs, ensuring even virtual SAN
utilization
C. Optimizes performance by consolidating all interswitch links into a single physical link
D. Provides the flexibility to deploy the same interswitch links for both FC and iSCSI traffic

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Question: What is a benefit of implementing the server flash caching technology in a storage environment?
A. Provides performance acceleration for read intensive workloads
B. Reduces the number of storage systems in a data center
C. Eliminates the need for cache on a storage system
D. Improves the network performance during peak workloads

Question: What does the transformation of an operating model for third platform adoption involve?

A. Provisioning IT resources through a self-service portal

B. Establishing new roles and responsibilities to manage IT services

C. Adopting virtualization technologies to enable automation

D. Building new technical and business skills in the IT staff

---

Question: What is an accurate statement about remote desktop services?

A. Operating system and applications hosted on a remote compute system are shared by multiple users

B. Users are provided dedicated virtual machines running operating system and applications, and hosted on a client system

C. Users have the flexibility to install applications and modify operating system settings on the remote compute system

D. Users' interface of the remote desktop is transmitted to the client without the use of a remote display protocol

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Question: What is an accurate statement about Infrastructure as a Service?

A. Consumer has control over the operating systems and the deployed applications

B. Consumer only has control over user-specific application configuration settings

C. Consumer has control over network devices, compute systems, and storage systems

D. Consumer only has control over the configuration settings of the application-hosting Environment

---

Question: How do VSANs ensure security in an IT infrastructure?

A. By providing isolation over a shared infrastructure

B. By authenticating a new switch that joins a fabric

C. By limiting the number of switches in a fabric

D. By prohibiting an unused switch port from being used

Question: What is a benefit of block-level storage virtualization?

A. Enables the online expansion of virtual volumes

B. Enhances performance by limiting the RSCN traffic

C. Improves security through logical segmentation of block storage devices

D. Increases scalability through separation of FC addresses across VSANs

---

Question: What is a benefit of application virtualization?

A. Avoids conflicts between different versions of the same application

B. Enables an application to directly access a LUN on a storage system

C. Runs applications without requiring the use of cache memory

D. Presents more memory to an application than is physically available

---

Question: What is a benefit of implementing a virtual machine (VM) storage migration?

A. Balances storage utilization by redistributing VMs across different storage systems

B. Enables VMs to keep functioning even if the source data is lost due to a failure

C. Improves the security of applications running inside VMs on the migrated storage system

D. Reduces the overall storage system capacity requirements for organizations

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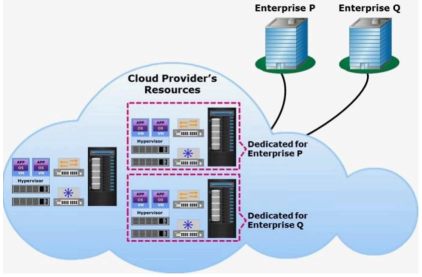
Question: A service provider is offering cloud services to Enterprise P and Enterprise Q.



Which cloud deployment model does the exhibit represent?

A. Externally-hosted private cloud

B. On-premise community cloud

C. Public cloud

D. Externally-hosted community cloud

Question: What is an accurate statement about a virtual desktop infrastructure?

A. Each user is provided a dedicated desktop VM with its own operating system and applications

B. Operating system and applications hosted on a client system are shared by multiple users

C. Each user has an individual desktop session on the remote compute system in which applications execute

D. Operating system and applications on the remote desktop are downloaded on the client for local execution

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Question: What is an accurate statement about infrastructure discovery?

A. Creates an inventory of infrastructure components and provides information about the components

B. Standardizes on-going infrastructure management activities and assesses potential risks of the activities

C. Classifies, organizes, and analyzes all infrastructure management best practices stored data lakes

D. Detects potential security attacks on a storage infrastructure and deploys specialized tools to eliminate the attacks

---

Question: What is signified by the "variability" characteristic of big data?

A. Meaning of the data changes constantly

B. Data is generated in different formats by numerous sources

C. Data change rate affects its timely analysis

D. Varying data quality affects its reliability and accuracy

Question: An administrator has defined a backup policy. Full backups will be performed at 8 PM every Monday and incremental backups performed at 8 PM the remaining days. How many backup copies are required to restore the files if they were accidentally deleted on Friday morning?

A. 1

B. 2

C. 3

D. 4

Question: What is an accurate statement about a virtual tape library?

A. Legacy backup software can be used without additional modules or changes

B. Portability of physical tapes tooffsite locations is supported

C. Only sequential data access for both backup and recovery is supported

D. Regular maintenance tasks associated with a physical tape drive are required

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Question: Which operation is performed by a scale-out NAS cluster when clients send data to the cluster?

A. Data is divided and stored on multiple nodes in parallel

B. Data is consolidated into a block and then stored on the NAS head

C. Data is evenly distributed to every file system in the cluster

D. NAS file system and cluster grow dynamically to store the data

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Question: Which technique protects the data in cache from power failures?

A. Vaulting

B. Mirroring

C. Cache tiering

D. Idle flushing

Question: What is an accurate statement about Software as a Service?

A. Consumer has limited control over user-specific application configuration settings

B. Consumer has control over the operating systems and deployed applications

C. Consumer has control over the configuration settings of the application-hosting environment

D. Consumer has limited control over select networking components, such as host firewalls

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Question: Which cloud service does VMware vCloud Air provide?

A. Infrastructure as a Service

B. Platform as a Service

C. Database as a Service

D. Software as a Service

Question: Which cloud computing characteristic enables multi-tenancy and abstracts the location of provided resources?

A. Resource pooling

B. Measured service

C. Rapid elasticity

D. Broad network access

Question: What is an accurate statement about the application streaming technique of application virtualization?

A. Application is downloaded in portions on the client for local execution

B. Application executes remotely and its user interface is transmitted to the client

C. Application executes on the client without the use of a locally-installed agent

D. Application is converted into a standalone package that directly executes on the client

---

Question: What is an accurate statement about the application presentation technique of application virtualization?

A. Application executes remotely and its user interface is transmitted to the client

B. Application is downloaded in portions on the client for local execution

C. Application executes on the client without the use of a locally-installed agent

D. Application is converted into a standalone package that directly executes on the client

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Question: In which data access method does the file system access an iSCSI storage system over a network?

A. Block-based

B. File-based
C. Object-based
D. Sector-based

Question: Which data access method enables a compute system to access a file system that resides on a storage system?

A. File-based
B. Block-based
C. Object-based
D. Sector-based

Question: What improves the performance of multi-threaded large block I/O workloads on a solid state drive?

A. Parallel I/O channels
B. I/O coalescing
C. Native command queuing
D. High capacity RAM cache

Question: An organization plans to deploy a new application in their environment. The new application requires 3 TB of storage space. During peak workloads, the
application is expected to generate 2450 IOPS with a typical I/O size of 2 KB. The capacity of each available disk drive is 500 GB. The maximum
number of IOPS a drive can perform at with 70 percent utilization is 90 IOPS.
What is the minimum number of disk drives needed to meet the application's capacity and performance requirements given a RAID 0 configuration?

A. 6
B. 16
C. 25
D. 28

Question: Which component of a software-defined storage (SDS) environment manages and provisions storage?

A. SDS controller
B. REST API
C. SDS web service
D. Storage agent

Question: Which data center characteristic ensures that data is stored and retrieved exactly as it was received?

A. Data integrity
B. Security
C. Scalability
D. Manageability

Question: What enables a logical unit to be presented to a host with more capacity than is physically allocated to it on the storage array?

a. Server Virtualization
b. Thin Provisioning
c. Storage Virtualization

d. File Virtualization

Question: What best describe external DAS?

a. Scalable Connectivity

b. Less protocol overhead

c. Ideal for file level access

d. Low complexity

Question: What is a function of block-level storage virtualization?

a. To abstract multiple physical networks as one logical network

b. To abstract multiple operating systems as one operating system

c. To abstract multiple physical arrays as a single target

d. To abstract multiple file servers as one file server

Question: What is a key benefit of virtual provisioning?

a. Reduced storage costs

b. Reduced network traffic

c. Reduced storage traffic

d. Support ILM strategy

Question: What is Disaster Recovery?

a. Estimating the amount of data loss after a disaster

b. Restarting mirror consistent copies of data and applications

c. Restoring a previous copy of data to a known point of consistency

d. Process required for an application to resume after a power failure

---

Question: How do Business Continuity and Disaster Recovery differ?

a. Business Continuity focuses on prevention; Disaster Recovery focuses on restoring business operations

b. Business Continuity focuses on prevention; Disaster Recovery focuses mitigating risks

c. Business Continuity is a part of Disaster Recovery process

d. Disaster Recovery involves planning, preparing for and recovering from an adverse event; Business Continuity focuses on supporting business operations after an adverse event

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Question: What action organizations need to take to secure the storage infrastructure?

a. Eliminate information assets

b. Repair vulnerabilities

c. Protect network access

d. Install threat agents

Question: What is the purpose of monitoring capacity in a storage infrastructure?

A. Tracking the amount of storage infrastructure resources used and available

B. Evaluating utilization of components and identifying bottlenecks

C. Tracking configuration changes to the storage infrastructure elements

D. Identifying component failures that may lead to service unavailability

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Question: What is an example of a security management task in a storage infrastructure?

A. Configuring LUN masking to restrict hosts from accessing a specific LUN

B. Enforcing storage space quotas to restrict users from overprovisioning resources
C. Restricting the number of simultaneous client access to application servers
D. Enabling sending messages to a support center in the event of resource failure

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Question: What is an example of an availability management task in storage infrastructure management?
A. Designing a new replication solution for data redundancy
B. Performing trend analysis of storage space consumption
C. Tracking configuration changes to storage components
D. Specifying storage space quotas for users

Question: What refers to an IT infrastructure's ability to function based on business expectations during its specified time of operations?
A. Information availability
B. Disaster recovery
C. Compliance adherence
D. Business agility

Question: What is a benefit of backing up data to a disk library?
A. Provides improved backup and recovery performance
B. Provides built-in portability to meet offsite requirements
C. Supports the shoe shining effect for improved backup performance
D. Pre-configured and dedicated appliance is only used for backups

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Question: Why is it important for organizations to maintain a copy of data at a remote site?
A. Ensure disaster recovery
B. Provide operational recovery
C. Ensure data archiving to meet regulatory requirements
D. Provide flexibility to use stub files for production operations

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Question: What is a benefit when a node is added to a scale-out NAS cluster?
A. Increased cluster performance
B. Reduced capital expenditure
C. Simplified cluster management
D. Decreased data duplication

Question: The IT department of an organization sends out an order to buy an additional 1 TB of storage based on the recommendation given by the data center manager. What could be the possible reason for this recommendation?
a. Deploy new monitoring tools
b. Implement clustering
c. Implement storage virtualization
d. Automate backup and recovery

Question: What is a reason for monitoring data center elements?
a. Ensure that no data is lost if there is disaster or service disruption
b. Ensure the IT support staffs are meeting all requirements of the service level agreements
c. Ensure the components are functioning properly and are available
d. Eliminate threats to storage infrastructure security

Question: What are two [2] key metrics for server performance?

a. Bandwidth utilization

b. Bus utilization

c. CPU utilization

d. Memory utilization

e. HBA/port utilization

Question: Which metric can be used as part of performance monitoring?

a. Login failures

b. Number of I/Os to disks in a storage array

c. Number of replication task failures

d. Number of used and free ports on different switches

Question: What is an advantage of a software-based object storage implementation?

A. Provides the flexibility to reuse the existing infrastructure

B. Requires less storage capacity to store a large number of objects

C. Provides the flexibility to create more than one object ID for an object

D. Requires less hardware to support scalable architecture